1) How many entries do you have in your database who have applied for Fall 2026?

```
SELECT COUNT(*)

FROM public.applicants

WHERE term = 'Fall 2026';
```

Explanation: This query counts all rows in public.applicants where term = 'Fall 2026'. It is used because each row represents one applicant entry, so COUNT(*) gives the total number of Fall 2026 applications.

2) What percentage of entries are from international students (not American or Other) (to two decimal places)?

```
SELECT ROUND(COUNT(*) / (SELECT COUNT(*) FROM public.applicants)::DECIMAL * 100.00,2) AS percentage_international

    FROM public.applicants

    WHERE us_or_international NOT IN ('American', 'Other')

      AND us_or_international IS NOT NULL;
```

Explanation: This query filters to rows where us_or_international is neither American nor Other, then divides that count by the total number of rows in the table and multiplies by 100. ROUND(..., 2) is used to format the final percentage to two decimal places.

3) What is the average GPA, GRE, GRE V, GRE AW of applicants who provide these metrics?

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS average_gpa,

  ROUND(AVG(gre)::numeric, 2) AS average_gre,

  ROUND(AVG(gre_v)::numeric, 2) AS average_gre_v,

  ROUND(AVG(gre_aw)::numeric, 2) AS average_gre_aw

  FROM public.applicants;
```

Explanation: This query uses AVG() to compute the mean of each numeric field across all rows. Since AVG() ignores NULL values, only applicants who actually provided each metric are included in that metric’s average. The results are cast and rounded to two decimal places for readability.:

4) What is the average GPA of American students in Fall 2026?

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS average_gpa
    FROM public.applicants
    WHERE us_or_international = 'American' AND term = 'Fall 2026';
```

Explanation: This query filters the table to applicants who are marked as American and whose term is Fall 2026, then computes the average gpa. It is the most direct way to isolate that subgroup and summarize their GPA.

5) What percent of entries for Fall 2026 are Acceptances (to two decimal places)?

```
SELECT ROUND(AVG(CASE WHEN status ILIKE '%Accept%' THEN 1 ELSE 0 END)::numeric * 100, 2) AS acceptance_percentage
    FROM public.applicants
    WHERE term = 'Fall 2026';
```

Explanation: This query limits the data to Fall 2026 applicants, then uses a CASE statement to convert accepted rows into 1 and all others into 0. Taking the average of those 1s and 0s gives the share of accepted applicants, and multiplying by 100 converts it to a percentage.

6) What is the average GPA of applicants who applied for Fall 2026 who are Acceptances?

```
SELECT ROUND(AVG(gpa)::numeric, 2) AS average_gpa
    FROM public.applicants
    WHERE term = 'Fall 2026' AND status ILIKE '%Accept%';
```

Explanation: This query filters to Fall 2026 rows where status indicates an acceptance, then calculates the average GPA for that subset. This is appropriate because it directly measures the average GPA among accepted Fall 2026 applicants only.

7) How many entries are from applicants who applied to JHU for a masters degrees in Computer Science?

```
SELECT COUNT(*)
    FROM public.applicants
```

```
    WHERE program ILIKE '%Johns Hopkins University%'
      AND degree ILIKE '%Masters%'
      AND program ILIKE '%Computer Science%';
```

Explanation: This query counts rows where the program text contains Johns Hopkins University, Computer Science, and the degree field contains Masters. It uses ILIKE so the text matching is case-insensitive and flexible for partial matches in the stored program string.

8) How many entries from 2026 are acceptances from applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University for a PhD in Computer Science?

```
SELECT COUNT(*)
    FROM public.applicants
    WHERE term = 'Fall 2026'
      AND status ILIKE '%Accept%'
      AND degree ILIKE '%PhD%'
      AND program ILIKE '%Computer Science%'
      AND (
        program ILIKE '%Georgetown University%' OR
        program ILIKE '%MIT%' OR
        program ILIKE '%Stanford University%' OR
        program ILIKE '%Carnegie Mellon University%'
      );
```

Explanation: This query counts Fall 2026 accepted applicants whose degree is PhD, whose program includes Computer Science, and whose program text matches one of the four target universities. The grouped OR conditions are used to capture any of those schools in a single query.

9) Do your numbers for question 8 change if you use LLM Generated Fields (rather than your downloaded fields)?

```
WITH CTE_LLM AS (
  SELECT COUNT(*) AS total_count_llm
   FROM public.applicants
   WHERE term = 'Fall 2026'
     AND status ILIKE '%Accept%'
     AND degree ILIKE '%PhD%'
     AND llm_generated_program ILIKE '%Computer Science%'
     AND (
       llm_generated_university ILIKE '%Georgetown University%' OR
       llm_generated_university ILIKE '%MIT%' OR
       llm_generated_university ILIKE '%Stanford University%' OR
       llm_generated_university ILIKE '%Carnegie Mellon University%')
     )

   ,CTE_DOWNLOADED AS (
   SELECT COUNT(*) AS total_count_downloaded
   FROM public.applicants
   WHERE term = 'Fall 2026'
     AND status ILIKE '%Accept%'
     AND degree ILIKE '%PhD%'
     AND program ILIKE '%Computer Science%'
     AND (
       program ILIKE '%Georgetown University%' OR
       program ILIKE '%MIT%' OR
```

```
        program ILIKE '%Stanford University%' OR
        program ILIKE '%Carnegie Mellon University%')
      )
  SELECT CASE WHEN total_count_llm = total_count_downloaded THEN 'No' ELSE 'Yes' END AS llm_discrepancy
  FROM CTE_LLM CROSS JOIN CTE_DOWNLOADED ;
```

Explanation: This query creates two counts: one using the LLM-generated fields (llm_generated_program, llm_generated_university) and one using the original downloaded program field. It then compares those counts and returns Yes if they differ and No if they are the same. This is useful for checking whether the normalized LLM-derived fields change the result.

10) Among Fall 2026 applicants, how does the acceptance rate differ by applicant type (American, International, Other)?

```
SELECT us_or_international,
  COUNT(*) AS total_applicants,
  COUNT(*) FILTER (WHERE status ILIKE 'Accepted%') AS accepted_applicants,
  ROUND(100.0 * COUNT(*) FILTER (WHERE status ILIKE 'Accepted%') / COUNT(*), 2) AS acceptance_rate_pct
  FROM public.applicants
  WHERE term = 'Fall 2026'
  AND us_or_international IS NOT NULL
  GROUP BY us_or_international
  ORDER BY acceptance_rate_pct DESC;
```

Explanation: This query groups applicants by us_or_international, counts the total applicants and accepted applicants in each group, and then calculates the acceptance rate as a percentage. Grouping by applicant type makes it possible to compare outcomes across categories in one result set.

11) For Computer Science applicants, which universities have the most acceptances (at least 20) in the dataset when using the LLM-generated university/program fields?

```
SELECT llm_generated_university,
  COUNT(*) AS acceptance_count
  FROM public.applicants
  WHERE status ILIKE 'Accept%'
  AND llm_generated_program ILIKE '%Computer Science%'
  AND llm_generated_university IS NOT NULL
  GROUP BY llm_generated_university
  HAVING COUNT(*) > 20
  ORDER BY acceptance_count DESC, llm_generated_university;
```

Explanation: This query filters to accepted Computer Science applicants using the LLM-generated program and university fields, groups the rows by llm_generated_university, and counts acceptances per school. The HAVING COUNT(*) > 20 clause keeps only universities with at least 21 acceptances, and the final ordering shows the highest-count schools first.